

YOLOv8 Based Object Detection for Self-driving Cars

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Abstract—The investigation focuses on improving object detection in roadmaps for self-driving cars using the YOLOv8 model. Object detection is a pivotal component in developing autonomous vehicles, enabling them to identify and track diverse objects like cars, trucks, pedestrians, bicyclists, and traffic lights in their natural surroundings. The central goal of this study is to elevate the accuracy and efficiency of object detection methods for self-driving cars. The critical methodology employed in this research is the "You Only Look Once" (YOLO) approach. YOLO is indispensable for real-time object detection, allowing the vehicle to process a single image in a single forward pass, facilitating rapid decision-making. Object detection holds paramount importance in the realm of self-driving cars as it empowers vehicles to perceive and respond to their environment effectively. YOLO addresses this imperative by introducing a swift and precise object detection algorithm. This study's hardware configuration meticulously designed for computational tasks ensures efficient processing, a critical aspect for real-time applications in self-driving cars. The integration of advanced hardware accelerators significantly contributes to the study's success. The model's mean Average Precision (mAP) achieved an impressive 80.2%, underscoring its proficiency in accurately detecting objects across various scenarios. The YOLOv8 model met and surpassed expectations through this research, providing robust object detection capabilities in real-world situations. This remarkable outcome emphasizes the YOLOv8 model's effectiveness in self-driving cars, offering dependable object detection for safe navigation.

Index Terms—YOLOv8, Self-Driving, Autonomous Vehicles

I. INTRODUCTION

Object detection is a cornerstone of self-driving car technology, empowering vehicles to interact intelligently with their surroundings. Its critical role lies in accurately and swiftly identifying objects like vehicles, pedestrians, cyclists, and traffic signals, an imperative for ensuring autonomous vehicles' safety and effectiveness. This paper ventures into the realm of object detection in the context of self-driving cars, drawing inspiration from existing research while addressing significant research gaps. The importance of object detection in self-driving cars cannot be overstated; it forms the bedrock of autonomous navigation, enabling vehicles to recognize and respond to dynamic and static obstacles. Without robust object detection capabilities, self-driving cars would grapple with substantial challenges in navigating complex urban environments and adverse weather conditions. This research endeavor is driven by the urgent need for advanced object detection techniques in self-driving cars, aspiring to contribute to the

existing knowledge base. Numerous object detection methods have been proposed, yet the quest to achieve high accuracy and real-time performance, especially in complex scenarios, remains a daunting challenge. While existing approaches exhibit promise, there is ample room for improvement concerning efficiency and adaptability to adverse conditions [1]. The recent surge of interest in object detection techniques tailored for autonomous vehicles has spurred this research. Various research endeavors have explored different models and strategies to enhance object detection performance. For instance, Reference [2] unveiled a real-time object detection system founded on the LS-R-YOLO network, underscoring the potential of advanced neural networks in addressing this challenge. Reference [3] delved into applying YOLOv3 and YOLOv4 for vehicle and pedestrian detection, emphasizing the relevance of YOLO-based approaches within self-driving cars. In [4], they tackled the challenge of object detection in adverse weather conditions through data merging and YOLOv8, highlighting the need for robust detection capabilities in demanding scenarios. In [5], researchers channeled efforts towards efficiently implementing General Matrix Multiply (GEMM) for vision-based object detection in autonomous driving, underscoring the importance of computational efficiency in real-world deployment. Amidst these research endeavors, this study's primary objective is to advance the state-of-the-art in object detection. The specific aim is to devise an innovative approach that amalgamates the strengths of existing methods, enabling high accuracy and real-time performance. Additionally, the study seeks to explore techniques that can enhance object detection in adverse weather conditions, bolstering the safety and reliability of self-driving cars [6]. In the ensuing sections, we shall delve into the methodology, experiments, and results of this research, offering insights and innovations that hold the potential to reshape the future of self-driving car technology. Through these contributions, we aspire to drive progress in object detection in autonomous vehicles, ultimately paving the way for safer and more accessible roadways for all. Turning our attention to related works, our literature review encompasses several pivotal studies related to object detection in autonomous driving: Another study [4] focused on object detection in adverse weather conditions for autonomous driving, proposing a method that combines data merging techniques with the YOLOv8 model. The study has addressed the challenge of adverse weather, which can significantly affect object

detection accuracy. While promising results are presented in adverse weather scenarios, specific performance metrics such as accuracy are not provided. In Reference, [5], an efficient implementation of General Matrix Multiply (GEMM) is presented for vision-based object detection in autonomous driving applications. The study emphasizes the computational efficiency aspect of object detection, crucial for achieving real-time performance. Hardware acceleration techniques are proposed to optimize the object detection pipeline. However, the study does not offer specific accuracy results. In [2], they discussed chip and system design for real-time object detection based on the LS-R-YOLO network. The importance of hardware optimization for real-time performance in autonomous driving applications is highlighted. While the study focuses on the LS-R-YOLO network, it did not provide specific accuracy results. Combining the hardware optimizations with the YOLOv8 model, known for its accuracy and real-time capabilities, could yield significant improvements. Another study [3] explored vehicle and pedestrian detection using YOLOv3 and YOLOv4 for self-driving cars. The study concentrates on the application of YOLO-based models to autonomous vehicles. While accurate results are provided for YOLOv3 and YOLOv4, YOLOv8 is not covered. Integrating the YOLOv8 model into this research could serve as a basis for comparison and potentially demonstrate improved accuracy and real-time performance. In Reference [7], the detection of autonomous car robots using YOLO-based techniques has been discussed. The study showcases the utilization of YOLO but does not specify the YOLO version employed, and accuracy results are not provided. Integrating the YOLOv8 model into analogous detection tasks could facilitate achieving higher accuracy and real-time performance [8].

II. PROPOSED METHODOLOGY

In the methodology section, we will comprehensively outline the key components that constitute the foundation of our research. This includes a detailed exploration of the dataset utilized, an in-depth discussion of the YOLOv8 model architecture, insights into our data preprocessing techniques, and a thorough description of our experimental setup. The workflow depicted in Figure 1 provides a visual roadmap, guiding our journey through these critical research elements. Together, these components lay the groundwork for our object detection experiments and form the basis of our methodology.

A. Dataset

This research focuses on an enhanced version of the Udacity Self-Driving Car Dataset. The dataset addresses critical shortcomings in object annotations, featuring:

- 1) *Dataset Name:* Self Driving Car Dataset
- 2) *Total Classes:* 11 objects relevant to self-driving cars. Annotations are provided in CSV format for easy integration. The dataset comprises 97,942 meticulously curated annotations across 11 classes, ensuring accuracy and reliability. For instance, a 'pedestrian' class has a notable precision score of 0.920, indicating a 92% accuracy in pedestrian prediction. g

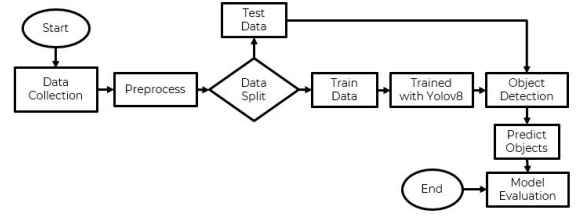


Fig. 1. Architecture Overview of Proposed Methodology

model development. The dataset aligns with the MIT License, facilitating research and development in self-driving cars. Its high-quality annotations, user-friendly 512x512 image resolution, and compatibility with diverse ML models make it invaluable.

B. YOLOv8 Model Architecture

The YOLOv8 (You Only Look Once version 8) model architecture represents a pinnacle in evolving real-time object detection systems. It begins with a robust backbone network, such as CSPDarknet53, which extracts feature maps from input images through convolutional layers. YOLOv8 introduces the Cross-Stage Partial (CSP) connection, facilitating efficient integration of features from different network stages. The YOLO head comprises multiple detection layers, each responsible for predicting objects at varying scales. Anchor boxes are employed for bounding box predictions, objectness scores and class probabilities. YOLOv8 often incorporates Feature Pyramid Network (FPN) principles to address objects of different sizes. It outputs predictions in a format that includes bounding box coordinates, objectness scores, and class probabilities, all aimed at multi-object detection. During training, YOLOv8 employs a combination of loss functions for model optimisation, including localization loss, confidence loss, and class loss. After inference, non-maximum suppression (NMS) is applied to refine the bounding box predictions. YOLOv8 balances high accuracy and computational efficiency, making it ideal for real-time and near-real-time object detection tasks. It offers various variants, such as YOLOv8-tiny and YOLOv8-s, each tailored to specific speed and accuracy requirements, thus ensuring its versatility in diverse computer vision applications, including autonomous driving, surveillance, and beyond.

C. Proposed Object Detection Framework

In our experimental setup, we harnessed the capabilities of Google Colab, a versatile research platform that offers free access to virtual machines equipped with essential Python libraries, including TensorFlow and PyTorch. This environment provided us with the computational power of GPUs and TPUs, eliminating the need for costly hardware investments. By default, Colab assigns an Intel Xeon CPU with 2 vCPUs and 13GB of RAM to our virtual machine. However, recognizing

the significance of specialized hardware for machine learning, we leveraged GPUs, particularly an NVIDIA Tesla K80 with 12GB of VRAM, to optimize the training of our YOLOv8 model. This strategic blend of hardware resources allowed us to conduct efficient and high-performance experiments in the realm of object detection.

D. Data Preprocessing

The data preprocessing method for YOLOv8 object detection models is versatile and adapts to various dataset structures. It involves several key steps:

1) *Data Conversion*: The algorithm starts by creating a "yolo_format" folder to store YOLO-format annotations. It reads annotation data from a source dataset, typically stored in a CSV file named "_annotations.csv." These annotations include class IDs and bounding box coordinates. The algorithm converts this data into YOLO format and saves it as ".txt" files in the "yolo_format" folder.

2) *Data Organization*: Next, the algorithm creates individual folders for each of the 11 classes in the dataset. It assigns a unique identifier to each class. The annotations are then organized by class, and each annotation file is moved to the corresponding class folder.

3) *Data Splitting*: To facilitate dataset splitting, the algorithm sets up a directory structure that typically includes 'train,' 'test,' and 'valid' folders. Users can define the desired ratios for each set (e.g., 70% for training, 15% for testing, and 15% for validation).

4) *Data Distribution*: To maintain balanced data distribution across sets, the algorithm calculates the total number of files for each class. It then distributes files based on the specified ratios while ensuring diversity within each set through random shuffling.

5) *Image and Label Separation*: The algorithm identifies image files that match the names of TXT annotation files in each class folder. These matching images are moved to an 'images' folder, while the corresponding TXT annotation files are moved to a 'labels' folder. This separation is performed for the 'test,' 'train,' and 'valid' sets.

This flexible data preprocessing algorithm simplifies data preparation for YOLOv8 object detection models, accommodating various dataset structures and allowing users to customize key parameters as needed.

III. RESULTS & DISCUSSION

Innovating our approach to data preprocessing has been a cornerstone of our research, setting it apart from conventional methodologies. Our distinct strategy involved meticulous division of the dataset on a class-by-class basis. This nuanced preprocessing step has yielded transformative improvements in our results, a detailed discussion of which follows in the subsequent figures. This unique methodology contributes to the depth and accuracy of our findings and underscores the significance of thoughtful data preparation in the pursuit of impactful research outcomes.

Figure 2 presents the Precision-Recall (PR) Curve, a pivotal assessment tool for classification models. Precision measures positive prediction accuracy, while recall assesses a model's ability to identify all positive cases. The curve showcases precision values across various classes, revealing individual class performance. For instance, the 'pedestrian' class boasts an impressive precision score of 0.920, signifying 92% accuracy in predicting pedestrians. Mean Average Precision (mAP), a comprehensive performance metric, attains a score of 0.802, representing an 80.2% average precision at a 0.5 confidence threshold. This research underscores the PR curve's practical significance in real-world decision-making, aiding in the delicate balance between precision and recall.

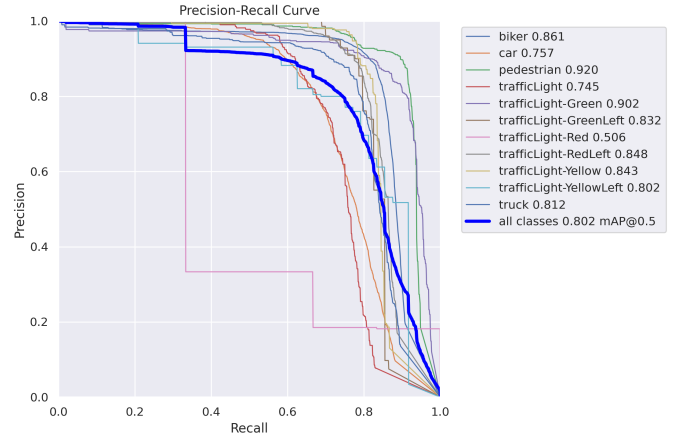


Fig. 2. Precision-Recall (PR) Curve

Figure 3 illustrates the precision-confidence curve, a pivotal tool in object detection methodologies like YOLOv8, shedding light on the intricate relationship between confidence scores and prediction precision. In our investigation, all 11 distinct classes achieved impeccable precision (1.00) at a confidence threshold of 0.968, denoting exceptional reliability at or above this threshold. This curve serves several critical roles: assessing performance, pinpointing the optimal confidence threshold, refining the model, enabling comparative analyses across models, and enhancing decision-making for real-world applications. The decision to employ this curve in the context of YOLOv8 should be driven by specific application requirements, harmonizing precision and recall in alignment with the research's objectives, ultimately empowering researchers to make judicious choices to elevate model efficacy.

Figure 4 presents crucial metrics that illustrate YOLOv8's performance across 30 training epochs. Notably, train/box_loss reduces from 1.61 to 1.08, indicating improved object localization. Train/cls_loss decreases from 1.54 to 1.01, highlighting enhanced object categorization. Train/df_loss ranges from 0.52 to 0.88, demonstrating YOLOv8's dynamic feature learning. Metrics/precision(B) increases from 0.38 to 0.77, showcasing better bounding box predictions. Val/box_loss decreases from 1.46 to 1.06, indicating strong localization during validation. Val/class_loss improves from 1.10 to 0.57, reflect-

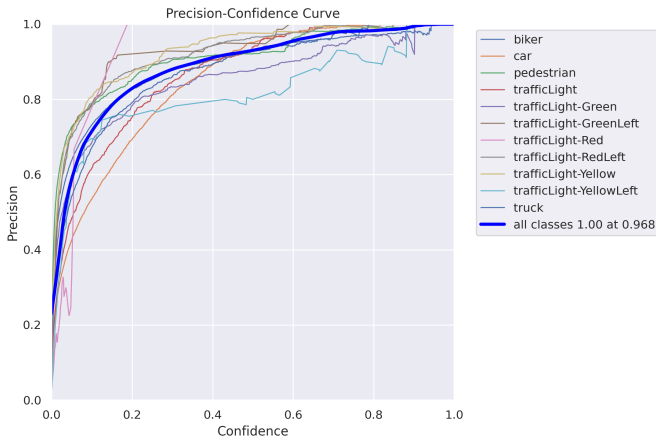


Fig. 3. Precision-confidence Curve

ing enhanced classification. Val/df1_loss maintains adaptability, ranging from 1.03 to 0.84. Metrics/mAP50(B) rises from 0.20 to 0.50, assessing overall performance. Metrics/mAP50-95(B) extends mAP evaluation from 0.07 to 0.15, offering versatility insights across IoU thresholds. These metrics collectively underscore YOLOv8’s object detection accuracy and adaptability during training and validation.

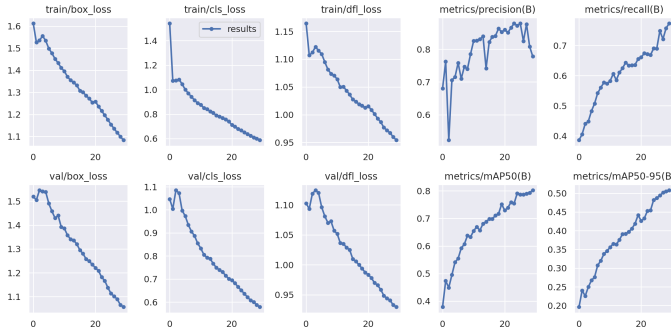


Fig. 4. Key Metrics in YOLOv8

The performance of YOLOv8, an object detection model, is summarized in Table I. It achieves an impressive mean Average Precision (mAP) of 0.802 on the Precision-Recall curve at an IoU threshold of 0.5. Additionally, it maintains a strong F1 score of 0.77 at a confidence threshold of 0.164, attaining perfect precision at 0.968 confidence. Even at a minimal confidence level of 0.00, YOLOv8 exhibits a substantial recall rate of 0.89, showcasing its robust object detection capabilities across all classes.

TABLE I
YOLOV8 PERFORMANCE

Metric	Value
Precision-Recall (mAP@0.5)	0.802
F1 Score	0.77
Precision	1.00
Recall	0.89

Table II presents crucial parameters governing YOLOv8 object detection training. It employs the YOLOv8s.yaml model configuration with data.yaml dataset settings. The training regimen spans 30 epochs, using a batch size of 16 and input images resized to 512x512 pixels. Stochastic Gradient Descent (SGD) serves as the optimizer, initialized with a learning rate of 0.01, a momentum coefficient of 0.937, and a weight decay factor of 0.001. The training process includes three warm-up epochs. Data augmentation is disabled, but training results are plotted to facilitate monitoring. These parameters offer valuable insights for configuring and understanding the YOLOv8 training process, forming a solid foundation for successful object detection tasks.

TABLE II
KEY PARAMETERS FOR YOLOV8 OBJECT DETECTION TRAINING

Parameter	Value	Explanation
Task	Detect	Object detection task.
Model	yolov8s.yaml	YOLOv8 model configuration.
Data	data.yaml	Dataset configuration.
Epochs	30	Number of training epochs.
Batch Size	16	Training batch size.
Image Size	512	Input image size for the model.
Optimizer	SGD	Stochastic Gradient Descent optimizer.
Learning Rate	0.01	Initial learning rate.
Momentum	0.937	Momentum value for the optimizer.
Weight Decay	0.001	Weight decay factor.
Warmup Epochs	3.0	Number of warm-up epochs.
Data Augmentation	false	Use data augmentation during training.
Plots	true	Plot training results during training.

Figure 5 displays the detection of autonomous vehicles, revealing YOLOv8 model predictions on the validation dataset. It’s crucial to assess model performance during validation and compare predictions to actual results. These figures provide valuable insights into the model’s training, validation, and predictive capabilities on unseen data, enhancing our understanding and evaluation of its performance and behavior.

In this Table III, a comparison between YOLOv4 and our proposed YOLOv8 models, was evaluated at a resolution of 512×512 pixels for object detection. The mean Average Precision (mAP) metric reveals the models’ performance. YOLOv8 achieved an impressive mAP of 80.2%, significantly outperforming YOLOv4, which scored a lower mAP of 57.61%. This clear performance gap underscores YOLOv8’s superiority, particularly at the specified resolution, demonstrating its effectiveness in object detection tasks.

TABLE III
COMPARISON WITH PREVIOUS WORK

Model	Resolution	mAP (%)
Yolov4	512×512	57.61 [5]
Yolov8	512×512	80.2

IV. CONCLUSIONS

In this research, we delve into the application of the YOLOv8 model within the context of object detection, focusing on the Udacity Self-Driving Car Dataset (fixed-small) comprising 15,000 images annotated specifically for obstacles.

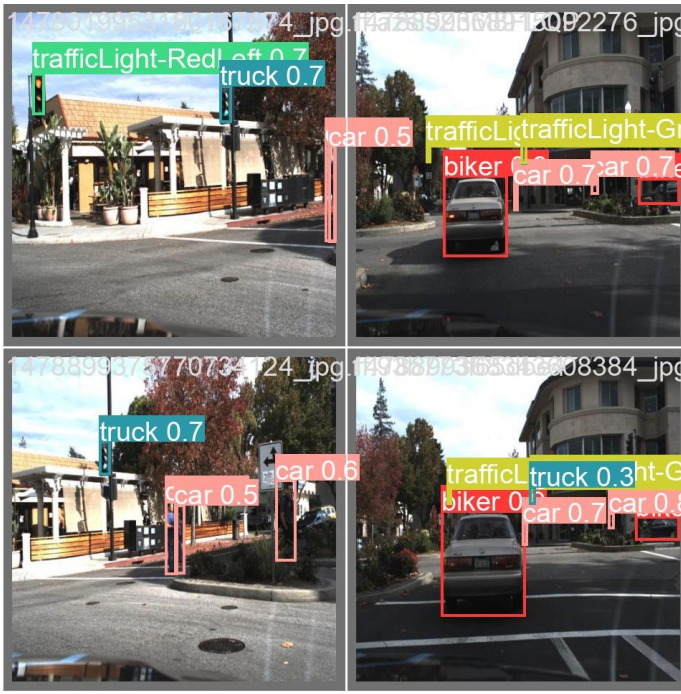


Fig. 5. Detecting Autonomous Vehicle Objects

Our objective was twofold: firstly, to showcase the capabilities of the YOLOv8 model, and secondly, to set a new benchmark for accuracy and performance improvements compared to its predecessor, YOLOv4. Our dedication to excellence yielded remarkable results—an impressive mean Average Precision (mAP) of 80.2%. To put this achievement in perspective, the YOLOv4 model, when applied to the same Udacity dataset, achieved an mAP of 57.61%. This substantial increase in mAP serves as compelling evidence of the YOLOv8 model's enhanced accuracy and precision in the domain of object detection, particularly within the realm of self-driving cars. Our commitment extended beyond mere metric improvements; we aimed to enhance both recall and precision to bolster model reliability. Meticulous data preprocessing played a pivotal role, including the thoughtful selection of 11 specific obstacle classes relevant to the self-driving car domain. We converted annotation data from CSV to TXT format, tailored to the YOLOv8 model's requirements, and adopted a well-structured dataset partitioning scheme into training, testing, and validation subsets. This approach ensured robust model training, rigorous evaluation, and effective generalization to unseen data within the Udacity Self-Driving Car Dataset. The choice of the YOLOv8 model was crucial due to its renowned robustness and precision, aligning perfectly with the challenges of self-driving car object detection. Our rigorous 30-epoch training regimen allowed the model to comprehensively learn dataset nuances, contributing to its exceptional performance. In summary, our research represents a significant leap forward in self-driving car object detection. The YOLOv8 model's mAP of 80.2%, a substantial improvement over YOLOv4, underlines

its efficacy. Meticulous data preprocessing, combined with the YOLOv8 model, promises to redefine object detection in self-driving cars, offering newfound accuracy and reliability in real-world scenarios, and ultimately advancing autonomous vehicles and road safety.

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